<< Title of the Project>>

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REPORT

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BONAFIDE CERTIFICATE

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ABSTRACT

TABLE OF CONTENTS

CHAPTER NO	TITLE	PAGE NO
	ABSTRACT	I
	LIST OF TABLES	II
	LIST OF FIGURES	III
	LIST OF ABBREVATIONS	IV
1	**INTRODUCTION**	1
	1.1 OBJECTIVE	1
	1.2 OVERVIEW	3
	1.3 SCOPE	4
2	**LITERATURE SURVEY**	9
3	**PROBLEM IDENTIFICATION**	19
4	PROPOSED SYSTEM	23
	4.1 SYSTEM ARCHITECTURE DIAGRAM	24
5	**RESULT AND DISCUSSION**	30
7	**CONCLUSION**	65
8	**REFERENCES**	66

LIST OF TABLES

TABLE NO	TITLE	PAGE NO
4.1	CELL SPECIFICATION	29
4.4.1	THERMOCHEMICAL PROPERTIES OF PARAFFIN WAX	34
4.5.1	THERMOCHEMICAL PROPERTIES OF PARAFFIN WAX	37
5.1	INSTRUMENT SPECIFICATIONS	50
6.1	THERMAL CONDUCTIVITY OF VARIOUS PCM SAMPLES	55

LIST OF FIGURES

FIG NO	TITLE	PAGE NO
1.2	TYPES OF PCM	5
3.1	FLOW CHART	19
4.1	DESIGN OF PHOTOVOLTAIC PANEL WITH FRONT VIEW AND SIDE VIEW	30
4.2	SOLAR PANEL	31

LIST OF ABBREVATIONS

PCM - PHASE CHANGE MATERIAL

PV- PHOTO VOLTAIC

DC-DIRECT CURRENT

AC-ALTERNATING CURRENT

BIPV- BUILDING INTEGRATED PHOTO VOLTAIC

LED-LIGHT EMITTING DIODE

CHAPTER - 1

INTRODUCTION

1.1 Introduction to Photovoltaic System

CHAPTER 2

LITERATURE SURVEY

CHAPTER -3

PROBLEM IDENTIFICATION AND METHODOLOGY

3.1 Problem Identification

CHAPTER 4

PROPOSED SYSTEM

RESULTS AND DISCUSSION

CONCLUSION

REFERENCES

1) Nelson J. The physics of solar cells. World Scientific Publishing Company; 2003

2) Shockley W, Queisser HJ. Detailed balance limit of efficiency of p-n junction solar cells. J App Phys 1961; 32:510–9.

3) Płaczek-Popko E. Top PV market solar cells 2016. Opto-Electron Rev 2017;25: 55–64.

4) Sharma A, Tyagi VV, Chen CR, Buddhi D. Review on thermal energy storage with phase change materials and applications. Renew Sustain Energy Rev 2009;13: 318–45.

5) Ma T, Zhao J, Li Z. Mathematical modelling and sensitivity analysis of solar photovoltaic panel integrated with phase change material. App Energy 2018; 228:1147–58.

6) Pandey A, Hossain M, Tyagi V, Rahim NA, Jeyraj A, Selvaraj L, et al. Novel approaches and recent developments on potential applications of phase change materials in solar energy. Renew Sustain Energy Rev 2018; 82:281–323.

7) Hausler, T. and Rogass, H. (2000) Latent Heat Storage. Sixteenth European Photovoltaic Solar Energy Conference, Glasgow, May 2000, 2265-2267.

8) Kibria M, Saidur R, Al-Sulaiman F, Aziz MMA. Development of a thermal model for a hybrid photovoltaic module and phase change materials storage integrated in buildings. Sol Energy 2016; 124:114–23.

9) Khanna S, Reddy K, Mallick TK. Performance analysis of tilted photovoltaic system integrated with phase change material under varying operating conditions. Energy 2017; 133:887–99.

1) Hasan A, McCormack SJ, Huang MJ, Norton B. Energy and cost saving of a photovoltaic-phase change materials (PV-PCM) system through temperature regulation and performance enhancement of photovoltaics. Energies 2014;7: 1318–31.
2) Ho C, Tanuwijava A, Lai C-M. Thermal and electrical performance of a BIPV integrated with a microencapsulated phase change material layer. Energy Build 2012; 50:331–8.
3) Ho C, Chou W-L, Lai C-M. Thermal and electrical performance of a water-surface floating PV integrated with a water-saturated MEPCM layer. Energy Convers Manage 2015; 89:862–72.